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Text Mining & NLP

COMM040

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Intended Learning Outcomes

- Understand the strengths, weaknesses and biases of different NLP methods
- Know about recent advances and applications of NLP methods, their impact and real world applications.
- Use modern NLP techniques to derive insight from unstructured data
- Communicate the pros and cons and trade-offs of different approaches to text analysis

Outline

Recap & Motivation

Semantic Properties

Semantic Vectors

PPMI

Predictive Models

Bias From Data

Outline



Recap & Motivation

Semantic Properties

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Predictive Models

Bias From Data

Last topics (summary)

- How to model different linguistic levels
- Various algorithms for representing and processing language
- Pros and cons of different methods

Motivation

- Lexical Semantics: lists are limited, resources lack coverage
- Introduction to automatic learning of meaning (useful for LLMs)

Classical Lexical Resources

- Limitations:
 - Human-built semantic resources
 - ◇ Example: WordNet
 - Limitations
 - ◇ Sparse coverage
 - ◇ Missing domain-specific vocabulary
 - ◇ Not quantitative
 - ◇ Poor scalability with language change

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Symbolic semantics

- How can we represent word meaning automatically?
- Properties? [menti.com](https://www.menti.com)

Semantic Similarity

- Semantic proximity example
 - “cat” closer to “dog” than to “economy”
- Semantic space: [menti.com](https://www.menti.com)

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The Distributional Hypothesis

- Foundational principle

“You shall know a word by the company it keeps.”

- Core idea
 - Words in similar contexts have similar meanings
- Importance
 - Foundation of modern embedding models

What Counts as Context?

- Window-based context (e.g., ± 5 words)
- Dependency-based context
 - Syntactic relations
- Document-level co-occurrence
- Skip-gram sampled context
- Consequence
 - Different context definitions produce different embeddings
 - Small windows capture functional similarity
 - Large windows capture topical similarity

How Context Choice Affects Semantics



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- Small context window (± 2)
 - Captures syntactic behaviour
 - Examples: run, walk, drive
- Large context window
 - Captures topical relations
 - Examples: economy, inflation, policy

Example



From Co-occurrence to Vectors

- Distributional vector definition
 - Summarises word co-occurrence behaviour
- Geometric intuition
 - Similar distributions imply spatial proximity
- Model families
 - Count-based models
 - Predictive models

The Co-occurrence Matrix

- Matrix structure
 - Rows: target words
 - Columns: context words
 - Entries: co-occurrence counts

[IMAGE PLACEHOLDER: small co-occurrence matrix example]

- Dimensionality
 - Vocabulary size V
 - Matrix size $V \times V$

Sparsity & Zipf's Law

- Sparsity
 - Most word pairs never co-occur
- Zipf's law
 - Few words are very frequent
 - Many words are extremely rare
- Consequence
 - Raw counts dominated by stopwords

Why Raw Counts Are Not Enough



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- Raw co-occurrence counts are not informative enough
- Example comparison
 - “cat” and “the”
 - ◇ High co-occurrence
 - ◇ Weak semantic relation
 - “cat” and “meow”
 - ◇ Rare co-occurrence
 - ◇ Strong semantic association
- Need for association measures
 - Capture meaningful semantic links
 - Go beyond frequency

Pointwise Mutual Information (PMI)



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- Purpose of PMI
 - Measures how much two words co-occur beyond chance
- $PMI(w, c) = \log \frac{P(w, c)}{P(w)P(c)}$
- Interpretation
 - **High PMI** Strong association (e.g., “cat”, “meow”)
 - **Low PMI** Weak association (e.g., “cat”, “stock”)
 - **Negative PMI** Co-occurrence lower than chance

Positive PMI (PPMI)

- Motivation
 - Negative PMI values are often uninformative
- Definition
 - $\text{PPMI}(w, c) = \max(\text{PMI}(w, c), 0)$
- Properties
 - Sparse representation
 - Emphasises semantically meaningful associations

Dimensionality Reduction via SVD



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- Problem
 - PPMI matrices are extremely large
- Solution
 - Apply Singular Value Decomposition (SVD)
- Effects
 - Reduced dimensionality
 - Typical k between 100 and 300
- Interpretation
 - Captures latent semantic structure
 - Foundation of Latent Semantic Analysis (LSA)

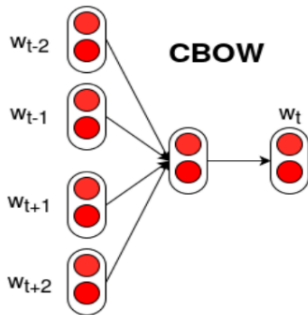
Why Predictive Models?

- Limitations of count-based models
 - Depend on global statistics
 - Require large matrices
 - Expensive SVD computations
- Predictive approach
 - Learn embeddings directly from prediction tasks
- word2vec
 - Marked a shift in semantic modelling

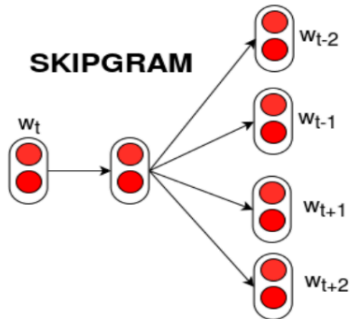
word2vec: Two Architectures

- CBOW (Continuous Bag-of-Words)
 - Predicts target word from surrounding context
- Skip-gram
 - Predicts surrounding context words from target word
 - More effective for rare words

word2vec Architectures



$$\frac{1}{T} \sum_{t=1}^T \sum_{-c \leq j \leq c, j \neq 0} \log p(w_t | w_{t+j})$$



$$\frac{1}{T} \sum_{t=1}^T \sum_{-c \leq j \leq c, j \neq 0} \log p(w_{t+j} | w_t)$$

Skip-Gram Model

...lemon, a [tablespoon of apricot jam, a] pinch...

c1 c2 [target] c3 c4

↑

positive examples +

t	c
apricot	tablespoon
apricot	of
apricot	jam
apricot	a

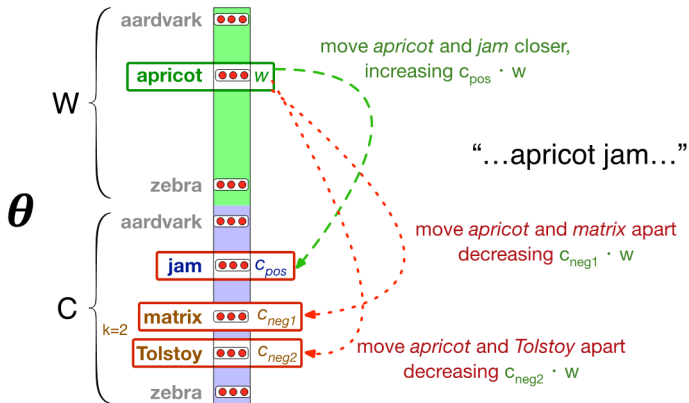
negative examples -

t	c	t	c
apricot	aardvark	apricot	seven
apricot	my	apricot	forever
apricot	where	apricot	dear
apricot	coaxial	apricot	if

Skip-Gram Model

- Negative Sampling
 - Key idea
 - ◇ Replace softmax with binary classification
 - Training samples
 - ◇ Positive samples: real (w, c) pairs
 - ◇ Negative samples: randomly drawn words
 - Objective
 - ◇ Distinguish real context pairs from noise
 - Benefit
 - ◇ Dramatically reduces computation

Skip-Gram Model

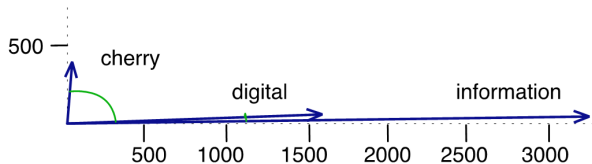


PPMI, word2vec, and Others

- Dense vectors
- Static embeddings

Vector Similarity

Dimension 1: 'pie'



Dimension 2: 'computer'

$$\text{cosine}(\vec{v}, \vec{w}) = \frac{\vec{v} \cdot \vec{w}}{|\vec{v}| |\vec{w}|} = \frac{\sum_{i=1}^N v_i w_i}{\sqrt{\sum_{i=1}^N v_i^2} \sqrt{\sum_{i=1}^N w_i^2}}$$

-1: opposite directions
+1: same directions
0: orthogonal vectors

What Do Static Embeddings Capture

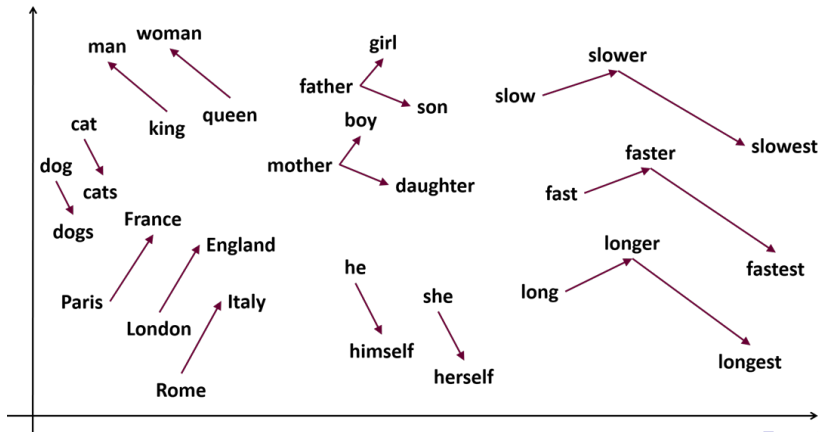


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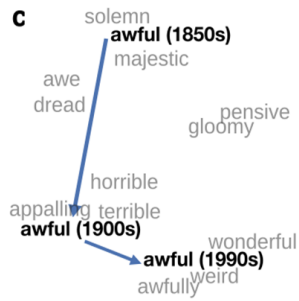
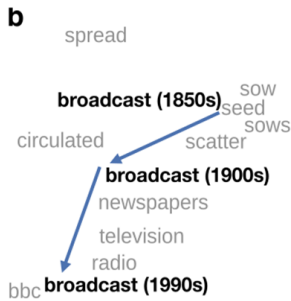
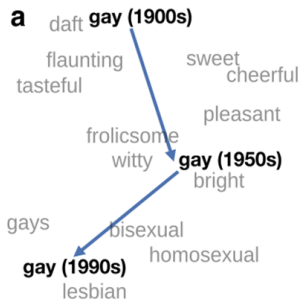
- Semantic similarity
 - cat $\leftrightarrow$ dog
 - Paris $\leftrightarrow$ London
- Syntactic similarity
 - run $\leftrightarrow$ walk
 - bigger $\leftrightarrow$ smaller
- Domain similarity
 - inflation $\leftrightarrow$ unemployment
- Linear relational structure

Linear Analogies

- king - man + woman $\approx$ queen



Semantic Shift



Nearest Neighbour Examples

- Example queries
 - “dog”
 - ◇ cat, puppy, pet, animal
 - “Paris”
 - ◇ France, Rome, Berlin
 - “doctor”
 - ◇ nurse, physician, surgeon

Limitations of Linear Analogies

- Common issues
 - Often cherry-picked
 - Inconsistent across corpora
 - Sensitive to frequency imbalance
 - Not robust to polysemy

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Gender Bias

- Simple game: What is the gender?
- [menti.com](https://www.menti.com)

Bias in Word Embeddings

- Source of bias
 - Patterns present in training data: social, historical, and cultural biases

Gender “man : computer programmer :: woman : homemaker”

Sentiment “immigrant” associated with negative neighbours in some corpora

- Representation

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